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United States Patent Application Publication

20250276312

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEM AND APPARATUS FOR REACTIONS INCLUDING A LIQUID TRANSFER DEVICE WITH AN ASYMMETRICAL CROSS-SECTION

Abstract

This disclosure provides systems, apparatuses, and methods for liquid transfer and performing reactions. In one aspect, a system includes a liquid transfer device having a housing having a pipette tip and a plunger assembly; and a reaction chamber, wherein the housing of the liquid transfer device is configured to sealably engage with the reaction chamber. In another aspect, a liquid transfer device including a housing having a pipette tip; and a plunger assembly disposed within the housing and the pipette tip, wherein a portion of the plunger assembly is configured to engage a fluid reservoir such that the plunger assembly remains stationary relative to the fluid reservoir and the housing moves relative to the plunger assembly.

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Family ID: 46889055

Appl. No.: 19/042544

Filed: January 31, 2025

Related U.S. Application Data

parent US continuation 17238841 20210423 parent-grant-document US 12269026 child US 19042544

parent US continuation 16057209 20180807 parent-grant-document US 11033894 child US

17238841

parent US continuation 13242999 20110923 parent-grant-document US 9352312 child US 15141190

parent US division 15141190 20160428 parent-grant-document US 10040061 child US 16057209

Publication Classification

Int. Cl.: **B01L3/02** (20060101); **A61J1/20** (20060101); **B01L3/00** (20060101)

U.S. Cl.:

CPC **B01L3/0217** (20130101); **B01L3/502** (20130101); A61J1/2096 (20130101); B01L2200/025 (20130101); B01L2200/026 (20130101); B01L2200/16 (20130101); B01L2300/025 (20130101); B01L2400/0478 (20130101); Y10T436/2575 (20150115)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/238,841, filed Apr. 23, 2021, now allowed, which is a continuation of U.S. patent application Ser. No. 16/057,209, filed Aug. 7, 2018, now U.S. Pat. No. 11,033,894, issued Jun. 15, 2021, which is a divisional of U.S. patent application Ser. No. 15/141,190, filed Apr. 28, 2016, now U.S. Pat. No. 10,040,061, issued Aug. 7, 2018, which is a continuation and claims priority to U.S. patent application Ser. No. 13/242,999, filed Sep. 23, 2011, now U.S. Pat. No. 9,352,312, issued May 31, 2016, the entire contents of each of which are incorporated by reference.

TECHNICAL FIELD

[0002] This invention relates to systems and apparatuses for liquid transfer and carrying out reactions.

BACKGROUND

[0003] Many diagnostic tests that involve biological reactions are required to be performed in laboratories by skilled technicians and/or complex equipment. Such laboratories may be the subject of government regulation. The costs of compliance with such regulations can increase the costs of diagnostic tests to patients and health care payers and exclude such tests from point-of-care facilities. There is a need for systems for performing diagnostic tests involving biological reactions that can be used without extensive training at the point of care.

SUMMARY

[0004] The present disclosure provides systems, apparatuses and methods for transfer of liquids and processing of reactions, e.g., in diagnostic tests.

[0005] In one aspect, the disclosure features a system that includes a liquid transfer device that includes a housing having a pipette tip and a plunger assembly; and a reaction chamber, wherein the housing of the liquid transfer device is configured to sealably engage with the reaction chamber. In some embodiments, the housing of the liquid transfer device can include a seal component configured to sealably engage with the reaction chamber. In some embodiments, the reaction chamber can include a seal component configured to sealably engage with the liquid transfer device. The systems can further include a fluid reservoir, and the reaction chamber can optionally be configured to lockably engage with the fluid reservoir.

[0006] The liquid transfer device can be configured to lockably engage with the reaction chamber, e.g., without dispensing, prior to dispensing, and/or after dispensing a liquid sample.

[0007] In some embodiments, the reaction chamber includes one or more components of a biological reaction.

[0008] In another aspect, the disclosure features a liquid transfer device that includes a housing having a pipette tip; and a plunger assembly disposed within the housing and the pipette tip, wherein a portion of the plunger assembly is configured to engage a fluid reservoir such that the plunger assembly remains stationary relative to the fluid reservoir and the housing moves relative to the plunger assembly.

[0009] In some embodiments, movement of the housing relative to the plunger assembly results in creation of a vacuum within the pipette tip and, optionally, the plunger assembly can be configured to lock in a position resulting in creation of the vacuum. The housing can be configured to move relative to the plunger assembly by pushing the housing down on the fluid reservoir. The device can further be configured to provide an auditory and/or visual indication that the plunger assembly is in a position resulting in the creation of the vacuum.

[0010] A system can include the liquid transfer device and one or more of a fluid reservoir and reaction chamber. When a reaction chamber is included, the reaction chamber can be configured to unlock the plunger assembly when the liquid transfer device and the reaction chamber are interfaced.

[0011] In another aspect, the disclosure features a liquid transfer device configured to draw a sample from a fluid reservoir by pushing the device against the reservoir and systems that include the liquid transfer device and one or both of a reaction chamber and fluid reservoir.

[0012] In the systems described above, two or all three of the liquid transfer device, reaction chamber, and fluid reservoir can have compatible asymmetric cross-sections.

[0013] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a sample reservoir using a liquid transfer device described above; and (ii) dispensing the liquid sample, e.g., into a reaction chamber comprising one or more components of a reaction.

[0014] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a fluid reservoir using a liquid transfer device (e.g., a liquid transfer device described above); and (ii) dispensing the liquid sample into a reaction chamber, wherein the liquid transfer device sealably engages with the reaction chamber during or prior to dispensing.

[0015] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a fluid reservoir using a liquid transfer device (e.g., a liquid transfer device described above); and (ii) dispensing the liquid sample into a reaction chamber, wherein the liquid transfer device lockably engages with the reaction chamber during or prior to dispensing. The methods can further include (iii) interfacing the reaction chamber and the fluid reservoir, such that the reaction chamber lockably engages with the fluid reservoir.

[0016] The systems, apparatuses, and methods disclosed herein can provide for simple analysis of unprocessed biological specimens. They can be used with minimal scientific and technical knowledge, and any knowledge required may be obtained through simple instruction. They can be used with minimal and limited experience. The systems and apparatuses allow for prepackaging or premeasuring of reagents, such that no special handling, precautions, or storage conditions are required. The operational steps can be either automatically executed or easily controlled, e.g., through the use of auditory and/or visual indicators of operation of the systems and apparatuses.

[0017] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0018] FIG. 1 is an exploded view of an exemplary system as described herein.

[0019] FIGS. 2A-2C are exploded views of system subassemblies.

[0020] FIG. 2D is a view of the system mated and joined.

[0021] FIGS. 3A-3D depict the system in use.

[0022] FIG. 4 depicts the system in the context of an exemplary detection device.

[0023] FIGS. 5A-5C depict the system in cross-section during sample collection.

[0024] FIGS. 6A-6D depict the system in cross-section during sample dispensing.

[0025] FIGS. 7A-7B depict single (7A) and double (7B) variants of the system.

DETAILED DESCRIPTION

[0026] This application describes systems, apparatuses, and methods for transfer of liquids and processing of biological reactions (e.g., nucleic acid amplification reactions).

[0027] Referring to FIG. 1, the system can include three subassemblies: a transfer device **100**, amplification chamber **200**, and an elution container **300**. Each subassembly can have a D-shaped or otherwise asymmetrical cross section **105**, **205**, **305** that is compatible with the other two subassemblies, such that the subassemblies may only be mated to each other in one orientation.

[0028] FIGS. 2A-2C, show exploded views of the subassemblies **100**, **200**, and **300**, respectively.

In FIG. 2A, the transfer device **100** includes a body **110** having a D-shaped or otherwise asymmetrical cross section **105** and a pipette tip **120**. The transfer device also includes a plunger unit **130** having a syringe plunger **135** that seals within the pipette tip **120** using an o-ring **140**. The plunger unit also includes flexible arms **131** having tabs **138** that are aligned with two sets of lower **112** and upper **113** slots in the body **110**. Ridges within the body **110** align with grooves in the plunger unit **130** to guide the plunger unit **130** up and down within the body **110**. When the plunger unit **130** is in the lower position, the tabs **138** insert into the lower slots **112**. When the plunger unit **130** is in the upper position, the tabs **138** insert into the upper slots **113**. A spring **150** fits over a spring guide **139** of the plunger unit **130**, and can be compressed against the cap **160** when the transfer device **100** is assembled. When the plunger unit **130** is in the upper position, an indicator **137** at the top of the spring guide **139** is visible through an indicator window **165** in the cap **160**.

[0029] In FIG. 2B, the amplification chamber **200** includes a body **210** having a D-shaped or otherwise asymmetrical cross-section **205** that is compatible with the cross-section **105** of the transfer device **100**. The amplification chamber body **210** also includes two tabs **215** that insert into either the lower slots **112** or upper slots **113** of the transfer device **100** when the two subassemblies are mated. The reaction chamber **200** also includes a microtube **220** having a retaining ring **225** that holds the microtube **220** within an aperture in the bottom of the amplification chamber body **210**. The microtube **220** can also have a seal **228** that covers the mouth **223** of the tube **220**. In some embodiments, the microtube **220** is optically permeable to allow monitoring of its contents. The amplification chamber **200** also includes a scaling component **230** that fits within the amplification chamber body **210** and over the microtube **220**, holding it in place. The scaling component **230** includes a pliant gasket **235** configured to seal against the pipette housing **180** when the two subassemblies are mated (see FIGS. 6A-6D). Two side tabs **240** are present near the bottom of the body **210** of the amplification chamber **200**.

[0030] In FIG. 2C, the elution container **300** has a D-shaped or otherwise asymmetrical cross-section **305** that is compatible with the cross-section **105** of the transfer device **100**. The elution container **300** includes an elution buffer reservoir **310** and a guide ring **320** compatible with a pipette housing **180** of the transfer device **100**. A seal can cover the mouth of the buffer reservoir **310** or guide ring **320**. Two notches **340** are present on the side walls **350** of the elution chamber **300**, into which insert the side tabs **240** of the amplification chamber **200** when the two subassemblies are mated.

[0031] FIG. 2D shows the three subassemblies of the system mated and joined for disposal. The

transfer device **100** locks into the amplification chamber **200** by insertion of the amplification chamber tabs **215** into the upper slots **113** of the transfer device **100**. Similarly, the amplification chamber **200** locks into the elution chamber **300** by insertion of the side tabs **240** of the amplification chamber **200** into the notches **340** of the elution chamber **300**. In this configuration, the patient sample and any amplified nucleic acids are sealed within the system to prevent contamination. Approximate dimensions of the joined system are shown.

[0032] FIGS. 3A-3D show an overview of the system in operation. In FIG. 3A, the transfer device **100** is positioned above the elution chamber **300** with their D-shaped cross-sections **105** and **305** aligned. In FIG. 3B, the transfer device **100** is pushed down on the elution chamber **300**, such that the pipette tip **120** enters the buffer reservoir **310** and the plunger unit **130** remains stationary relative to the body **110** due to contact with a guide ring on the buffer reservoir **310**. This results in the plunger unit **130** in the upper position, compressing the spring **150** such that the indicator **137** shows through the indicator window **165**. The presence of the indicator **137** in the indicator window **165** and an audible click as the tabs **138** insert into the upper slots **113** provide auditory and visual feedback that the transfer device has been manipulated properly such that the pipette tip **120** is able to withdraw a portion of the sample from the buffer reservoir **310**. In FIG. 3C, the transfer device **100** has been removed from the elution chamber **300** and positioned above the amplification chamber **200** with their D-shaped cross-sections **105** and **205** aligned. In FIG. 3D, the transfer device **100** is pushed onto the amplification chamber **200**. The two tabs **215** of the amplification chamber **200** insert into the upper slots **113** of the transfer device **100**, displacing the tabs **138** and allowing the compressed spring **150** to relax and the plunger unit **130** to return to the lower position. The indicator **137** is no longer visible in the indicator window **165**, signaling that the contents of the pipette tip **120** have been emptied into the microtube **220**. The transfer device **100** is locked into the amplification chamber **200** by insertion of the amplification chamber tabs **215** into the upper slots **113** of the transfer device **100**.

[0033] FIG. 4 shows the system with an exemplary detection device **400**. The detection device **400** includes a first station **410** adapted to securely hold the elution chamber **300** and a second station **420** adapted to securely hold the amplification chamber **200**. When in use, the transfer device **100** is moved between the elution chamber **300** at the first station **410** and the amplification chamber **200** at the second station **420**. The detection device includes a lid **430** that can be closed when the detection device **400** is in operation or for storage. A touchscreen user interface **440** is present for inputting data and displaying information regarding the assay. The second station **420** can include a bar code reader or similar device to automatically detect a bar code or similar code present on the amplification chamber **200**. The first **410** and second **420** stations can be adapted to heat or cool the contents of the elution chamber **300** and reaction chamber **200**. The second station **420** can also be adapted to provide optical, fluorescence, or other monitoring and/or agitation of the microtube **220**.

[0034] FIGS. 5A-5C show the system in cross-section during sample collection. In FIG. 5A, the transfer device **100** is placed above the elution chamber **300** such that their cross sections **105**, **305** are aligned. The plunger unit **130** is in the lower position and the tabs **138** are in the lower slots **112**. In FIG. 5B, the transfer device **100** is lowered until one or more flanges **139** on the lower surface of the plunger unit **130** contact the guide ring **320**, and the pipette tip **120** and plunger tip **132** are inserted into the liquid sample **360**. The liquid sample **360** can be a patient or other sample or include a patient or other sample dissolved or suspended in a buffer. In FIG. 5C, the transfer device **100** is pushed down by the user into the elution chamber **300**. The plunger unit **130** remains stationary through the contact of the one or more flanges **139** against the guide ring **320**, while the transfer device body **110** is lowered relative to the plunger unit **130** and elution chamber **300**. Simultaneously, a guide channel **116** in the transfer device is pushed downward relative to the guide ring **320**. The downward motion of the transfer device body **110** causes the pipette tip **120** to move downward relative to the plunger tip **132** and draw a liquid sample portion **365** into the pipette tip **120**. The downward motion of the transfer device body **110** relative to the plunger unit

130 also compresses the spring **150**, moves the tabs **138** from the lower slots **112** to the upper slots **113**, and causes the indicator **137** to be visible through the indicator window **165**. The transfer device **100** with the liquid sample portion **365** can now be lifted off of the elution chamber **300** and is ready for transfer and dispensing.

[0035] FIGS. **6A-6D** show the system in cross-section during sample dispensing. In FIG. **6A**, the transfer device **100** is placed above the amplification chamber **200** such that their cross sections **105**, **205** are aligned. The amplification chamber **200** is held within the second station **420** of the detection device **400** with the microtube **220** seated within a tube holder **428**. In FIG. **6B**, the transfer device **100** is lowered until two inner tabs **250** within the amplification chamber **200** engage two ridges **170** in the lower sides of the transfer device body **110**, the tabs **215** insert into the lower slots **112** of the transfer device **100**, and the gasket **235** engages the pipette housing **180**. This prevents the transfer device **100** from being easily removed from the amplification chamber **200** once dispensing has been started and prevents release of the sample. In FIG. **6C**, the transfer device **100** is further lowered onto the amplification chamber **200**, such that the amplification chamber tabs **215** insert into the upper slots **113** of the transfer device and displace the plunger unit tabs **138**. Simultaneously, the pipette tip **120** pierces the seal **228** on the microtube **220**. In FIG. **6D**, the plunger unit **130**, no longer held in the upper position, moves to the lower position as the spring **150** expands. This causes the plunger tip **132** to move downward within the pipette tip **120** and dispense the liquid sample portion **365** into the microtube **220**. The liquid sample portion **365** rehydrates a dried reagent pellet **280** in the microtube **220**, initiating reaction (e.g., an amplification reaction). The transfer device **100** is locked in place on the amplification chamber **200** by the tabs **215** inserted into the upper slots **113**, and any product of the amplification reaction is sealed within the unit by the gasket **235**.

[0036] FIGS. **7A** and **7B** are three-quarter cross sections showing the system configured for one or two microtubes **220**. FIG. **7A** shows the transfer device **100** and amplification chamber **200** as described above with one pipette tip **120** and one microtube **220**. FIG. **7B** shows the transfer device **100** and amplification chamber **200** with two pipette tips **120** and two microtubes **220**. Using the device in FIG. **7B**, parallel reactions (e.g., amplification reactions) can be performed on two portions of one sample.

[0037] The systems and apparatuses disclosed herein can be used to perform reactions, e.g., utilizing biological components. In some embodiments, the reactions involve production of nucleic acids, such as in nucleic acid amplification reactions. Exemplary nucleic acid amplification reactions suitable for use with the disclosed apparatuses and systems include isothermal nucleic acid amplification reactions, e.g., strand displacement amplification, nicking and extension amplification reaction (NEAR) (see, e.g., US 2009/0081670), and recombinase polymerase amplification (RPA) (see, e.g., U.S. Pat. Nos. 7,270,981; 7,666,598). In some embodiments, a microtube can contain one or more reagents or biological components, e.g., in dried form (see, e.g., WO 2010/141940), for carrying out a reaction.

[0038] The systems and apparatuses disclosed herein can be used to process various samples in reactions, e.g., utilizing biological components. In some embodiments, the samples can include biological samples, patient samples, veterinary samples, or environmental samples. The reaction can be used to detect or monitor the existence or quantity of a specific target in the sample. In some embodiments, a portion of the sample is transferred using a transfer device as disclosed herein.

[0039] In some embodiments, a liquid transfer device or pipette tip disclosed herein can be configured to collect and dispense a volume between 1 μ l and 5 ml (e.g., between any two of 1 μ l, 2 μ l, 5 μ l, 10 μ l, 20 μ l, 50 μ l, 100 μ l, 200 μ l, 500 μ l, 1 ml, 2 ml, and 5 ml).

[0040] The disclosure also features articles of manufacture (e.g., kits) that include one or more systems or apparatuses disclosed herein and one or more reagents for carrying out a reaction (e.g., a nucleic acid amplification reaction).

[0041] A number of embodiments of the invention have been described. Nevertheless, it will be

understood that various modifications may be made without departing from the spirit and scope of the invention. For example, a transfer device as described herein can include three or more pipette tips. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A liquid transfer system comprising: a liquid transfer device comprising a housing and at least one pipette tip; a reaction chamber configured to sealably engage with the liquid transfer device; and a fluid reservoir configured to engage with at least one of the liquid transfer device or the reaction chamber, wherein at least two of the liquid transfer device, reaction chamber, and fluid reservoir have compatible asymmetric cross-sections.
2. The system of claim 1, wherein the liquid transfer device further comprises a plunger assembly disposed within the housing.
3. The system of claim 2, wherein the plunger assembly is configured to remain stationary relative to the fluid reservoir when the liquid transfer device engages with the fluid reservoir.
4. The system of claim 1, wherein the liquid transfer device is configured to lockably engage with the reaction chamber.
5. The system of claim 1, wherein the reaction chamber is configured to lockably engage with the fluid reservoir.
6. The system of claim 1, wherein the reaction chamber contains one or more components of a biological reaction.
7. The system of claim 1, wherein the liquid transfer device is configured to provide an auditory or visual indication when a sample has been drawn into the at least one pipette tip.
8. A method of transferring a liquid sample, comprising: obtaining a liquid sample from a fluid reservoir using a liquid transfer device; engaging the liquid transfer device with a reaction chamber; and dispensing the liquid sample into the reaction chamber, wherein at least two of the liquid transfer device, reaction chamber, and fluid reservoir have compatible asymmetric cross-sections.
9. The method of claim 8, further comprising lockably engaging the liquid transfer device with the reaction chamber prior to or during dispensing.
10. The method of claim 8, further comprising interfacing the reaction chamber with the fluid reservoir after dispensing the liquid sample.
11. A liquid transfer device comprising: a housing having at least one pipette tip; a plunger assembly disposed within the housing; and an asymmetric cross-section configured to mate with a complementary asymmetric cross-section of a separate component.
12. The liquid transfer device of claim 11, wherein the plunger assembly is configured to lock in a position that creates a vacuum within the at least one pipette tip.
13. The liquid transfer device of claim 11, further comprising an indicator configured to provide a visual or auditory signal when a sample has been drawn into the at least one pipette tip.
14. A system for performing a biological reaction, comprising: a liquid transfer device; a reaction chamber containing one or more biological reagents; and a fluid reservoir, wherein the system is configured to allow transfer of a sample from the fluid reservoir to the reaction chamber using the liquid transfer device, and wherein at least two components of the system have compatible asymmetric cross-sections to ensure proper alignment during use.
15. The system of claim 14, wherein the reaction chamber is configured to initiate a biological reaction upon receiving the sample from the liquid transfer device.
16. The system of claim 1, wherein the at least one pipette tip comprises a plurality of pipette tips.
17. The system of claim 16, wherein the plunger assembly comprises a plurality of syringe plungers, each syringe plunger operably connected to a corresponding pipette tip of the plurality of pipette tips.

18. The system of claim 1, wherein the fluid reservoir comprises an inner wall and an outer wall, and wherein the liquid transfer device is configured to engage with the fluid reservoir between the inner wall and the outer wall.

19. A method of preparing a biological reaction, comprising: obtaining a sample using a liquid transfer device from a fluid reservoir; transferring the sample to a reaction chamber containing dried biological reagents; sealing the reaction chamber with the liquid transfer device; and initiating a biological reaction within the sealed reaction chamber.

20. The method of claim 19, further comprising detecting a product of the biological reaction while the reaction chamber remains sealed by the liquid transfer device.
